

*** Solutions ***

1. (12 pts) Find the equation of the tangent plane to $f(x, y) = \sqrt{\frac{x}{y}}$ at the point $(81, 4)$.

Answer: $z = \frac{9}{2} + \frac{1}{36}(x-81) - \frac{9}{16}(y-4)$

Show work here:

$$f(x, y) = \left(\frac{x}{y}\right)^{1/2}$$

$$f(81, 4) = \sqrt{\frac{81}{4}} = \frac{9}{2}$$

$$f_x = \frac{1}{2} \left(\frac{x}{y}\right)^{-1/2} \left(\frac{1}{y}\right) = \frac{1}{2y\sqrt{\frac{x}{y}}} = f_x(81, 4) = \frac{1}{48 \cdot \frac{9}{2}} = \frac{1}{36}$$

$$f_y = \frac{1}{2} \left(\frac{x}{y}\right)^{-1/2} \left(-\frac{x}{y^2}\right) = \frac{-x}{2y^2\sqrt{\frac{x}{y}}} \Rightarrow f_y(81, 4) = \frac{-81 \cdot 9}{16 \cdot 36 \cdot \frac{9}{2}} = -\frac{9}{16}$$

2. (9 pts) Estimate $\Delta f = f(2.08, 3.92) - f(2, 4)$ assuming that $f(2, 4) = 5$, $f_x(2, 4) = 0.3$, $f_y(2, 4) = -0.2$. Simplify your answer.

Answer: $\Delta f \approx 0.04$

Show work here:

$$f(2.08, 3.92) \approx L(2.08, 3.92)$$

$$\begin{aligned} L(2.08, 3.92) &= 5 + 0.3(2.08 - 2) - 0.2(3.92 - 4) \\ &= 5 + 0.3(0.08) - 0.2(-0.08) \\ &= 5 + 0.024 + 0.016 \\ &= 5.04 \end{aligned}$$

$$\Delta f = f(2.08, 3.92) - f(2, 4) \approx L(2.08, 3.92) - f(2, 4) = 5.04 - 5 = 0.04$$

OR use

$$\Delta f = f(x, y) - f(a, b) \approx f_x(a, b) \Delta x + f_y(a, b) \Delta y$$

3. (12 pts) Let $f(x, y, z) = 3y^2 - xy + z^2$, $\mathbf{v} = \langle 4, -3, 0 \rangle$, and $P = (-2, 1, 2)$.

(3.1) a. A unit vector that points in the direction of the maximum rate of increase f at P is

$$\underline{\langle -1/9, 8/9, 4/9 \rangle}.$$

b. The maximum rate of increase of f at P is 9.

Show all work:

$$\nabla f = \langle -y, -x + 6y, 2z \rangle$$

$$\nabla f_P = \nabla f(-2, 1, 2) = \langle -1, 8, 4 \rangle$$

$$\|\nabla f(P)\| = \sqrt{(-1)^2 + 8^2 + 4^2} = \sqrt{1 + 64 + 16} = \sqrt{81} = 9$$

(3.2) Find the directional derivative of f in the direction of $\mathbf{v}$ at the point P . Fill in the bubble that corresponds to your answer. No credit will be given if more than one answer is selected.

(A) $\langle 0, 0, 0 \rangle$

(E) $\langle -2, 1, 2 \rangle$

(I) -28

(B) 5

(F) 9

(J) $\langle 4, -3, 0 \rangle$

(C) $-\frac{28}{9}$

(G) $\langle -1, 8, 4 \rangle$

(K) $\left\langle -\frac{4}{5}, -\frac{24}{5}, 0 \right\rangle$

(D) $\left\langle -\frac{4}{25}, -\frac{24}{25}, 0 \right\rangle$

(H) $-\frac{28}{25}$

☒ (L) $-\frac{28}{5}$

$$\vec{u} = \frac{\vec{v}}{\|\vec{v}\|} = \frac{\langle 4, -3, 0 \rangle}{\sqrt{16+9+0}} = \langle 4/5, -3/5, 0 \rangle$$

$$D_{\vec{u}} f(P) = \langle -1, 8, 4 \rangle \cdot \langle 4/5, -3/5, 0 \rangle = -\frac{4}{5} - \frac{24}{5} + 0 = -\frac{28}{5}$$

4. (9 pts) Let $f(x, y)$ be such that $\nabla f(x, y) = \langle y^4, 4xy^3 \rangle$, and let $x(s, t) = s^2t$ and $y(s, t) = t^2$. Apply the multivariable calculus chain rule to calculate $\frac{\partial f}{\partial t}$. Write only your final answer in the box provided. Your final answer must be simplified and written in terms of s and t .

$$\frac{\partial f}{\partial t} = \boxed{9s^2t^8}$$

Show your work here:

$$\frac{\partial x}{\partial t} = s^2, \quad \frac{\partial y}{\partial t} = 2t$$

$$\frac{\partial f}{\partial t} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial t}$$

$$= y^4 (s^2) + 4xy^3 (2t)$$

$$= (t^2)^4 (s^2) + 4 (s^2t) (t^2)^3 (2t)$$

$$= s^2t^8 + 8s^2t^8$$

5. (6 pts) Use implicit differentiation to find $\frac{\partial z}{\partial x}$ for $z^3 + zx + x^2y = x + 2yz$. Fill in **one** bubble that corresponds to your answer. Questions with multiple answers selected will receive no credit. No work required.

(A) $z + 2xy - 1$

(E) $-\frac{3z^2 + x - 2y}{z + 2xy - 1}$

☒ (I) $-\frac{z + 2xy - 1}{3z^2 + x - 2y}$

(B) $3z^2 + x - 2y$

(F) $x^2 - 2z$

(J) $-\frac{z + 2xy}{3z^2 + x}$

(C) $\frac{z + 2xy - 1}{3z^2 + x - 2y}$

(G) $-\frac{3z^2 + x - 2y}{x^2 - 2z}$

(K) $-3z^2 - x + 2y$

(D) $-\frac{x^2}{3z^2 + x}$

(H) $\frac{x^2 - 2z}{3z^2 + x - 2y}$

(L) $-\frac{x^2 - 2z}{3z^2 + x - 2y}$

$$F(x, y, z) = z^3 + zx + x^2y - x - 2yz = 0$$

$$F_x = z + 2xy - 1$$

$$F_z = 3z^2 + x - 2y$$

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} = -\frac{z + 2xy - 1}{3z^2 + x - 2y}$$

6. (10 pts) Find the critical point(s) of $f(x, y) = x^3 + 3xy - y^3$. Write your final answer(s) on the line provided.

Critical point(s): $(0, 0), (1, -1)$

Show work here:

$$\begin{cases} (1) f_x = 3x^2 + 3y = 0 \Rightarrow 3y = -3x^2 \Rightarrow y = -x^2 \\ (2) f_y = 3x - 3y^2 = 0 \end{cases}$$

Substitute into $y = -x^2$ (2): $3x - 3(-x^2)^2 = 0$

$$\Rightarrow 3x - 3x^4 = 0$$

$$\Rightarrow 3x(1 - x^3) = 0$$

$$\Rightarrow x = 0, x = 1$$

When $x = 0$ from (1) $y = -0^2 = 0$, so $(0, 0)$ is a critical pt

When $x = 1$ from (1) $y = -1^2 = -1$, so $(1, -1)$ is a critical pt

7. (10 pts) Let $P = (1, 1)$ be a critical point of a function $f(x, y)$, and let $Q = (0, 2)$ be a critical point of a function $g(x, y)$. Assume that $f_{xx}(P) = 4$, $f_{xy}(P) = 3$, $f_{yy}(P) = 2$ and $g_{xx}(Q) = -1$, $g_{xy}(Q) = 2$, $g_{yy}(Q) = -8$.

(7.1) Apply the second derivative test to classify P . Fill in the bubble that corresponds to your answer. No credit will be given if more than one answer is selected.

☐ (A) f has a local maximum at P .

☐ (B) f has a absolute maximum at P .

☐ (C) f has a local minimum at P .

☐ (D) f has a absolute minimum at P .

☒ (E) f has a saddle point at P .

☐ (F) Second derivative test is inconclusive at P .

Provide a complete justification for your answer using the Second Derivative Test below:

$$D(P) = \begin{vmatrix} 4 & 3 \\ 3 & 2 \end{vmatrix} = 8 - 9 = -1 < 0$$

must include

(7.2) Apply the second derivative test to classify Q . Fill in the bubble that corresponds to your answer. No credit will be given if more than one answer is selected.

☒ (A) g has a local maximum at Q .

☐ (B) g has an absolute maximum at Q .

☐ (C) g has a local minimum at Q .

☐ (D) g has a absolute minimum at Q .

☐ (E) g has a saddle point at Q .

☐ (F) Second derivative test is inconclusive at Q .

Provide a complete justification for your answer using the Second Derivative Test below:

$$D(Q) = \begin{vmatrix} -1 & 2 \\ 2 & -8 \end{vmatrix} = 8 - 4 = 4 > 0$$

must include

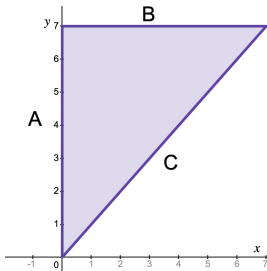
$$\text{and } g_{xx}(Q) = -1 < 0$$

must include

$$\text{or } g_{yy}(Q) = -8 < 0$$

8. (12 pts) Let $f(x, y) = x^2 + y^2 - 4x - 10y$ on the domain D such that $x \geq 0$, $0 \leq y \leq 7$, and $y \geq x$ (see figure below).

- (8.1) The maximum and minimum values of f on edge A have been provided. Find the maximum and minimum values of f on each of the remaining two edges of the triangle for the domain D . Complete the table with your final answers.



edge of triangle	A	B	C
max value of f	0	0	0
min value of f	-25	-25	$-\frac{49}{2}$

Show work below. You must provide justification for your answers for each remaining edge.

$$B: y=7, 0 \leq x \leq 7 : f(x, 7) = x^2 + 49 - 4x - 70 = x^2 - 4x - 21$$

$$\text{Let } g(x) = x^2 - 4x - 21$$

$$g'(x) = 2x - 4 = 0 \Rightarrow x = 2$$

$$\underbrace{g(0)}_{=f(0,7)} = -21, \quad \underbrace{g(2)}_{=f(2,7)} = 2^2 - 4(2) - 21 = -25, \quad \underbrace{g(7)}_{=f(7,7)} = 7^2 - 4(7) - 21 = 0$$

$$C: y=x, 0 \leq x \leq 7 : f(x, x) = x^2 + x^2 - 4x - 10x = 2x^2 - 14x$$

$$\text{Let } h(x) = 2x^2 - 14x$$

$$h'(x) = 4x - 14 = 0 \Rightarrow x = 7/2$$

$$\underbrace{h(0)}_{=f(0,0)} = 0, \quad \underbrace{h(7/2)}_{=f(7/2, 7/2)} = 2\left(\frac{7}{2}\right)^2 - 14\left(\frac{7}{2}\right) = -\frac{49}{2}, \quad \underbrace{h(7)}_{=f(7,7)} = 2(7)^2 - 14(7) = 0$$

- (8.2) The only critical point of the function f is $(2, 5)$. Determine the absolute maximum and absolute minimum of value of f on the domain D . Put your answers in the spaces provided.

The absolute max value of f is 0 and the absolute min value of f is -29.

Show any additional work here:

$$\begin{aligned} f(2, 5) &= 2^2 + 5^2 - 4(2) - 10(5) \\ &= 4 + 25 - 8 - 50 \\ &= -29 \end{aligned}$$

9. (10 pts) Consider the problem of finding the point on the ellipse $x^2 + 2y^2 + 2xy = 8$ with the greatest x -coordinate. Fill in the blanks below to set up this problem using the method of Lagrange multipliers.

(9.1) The function you need to maximize is $f(x, y) = \underline{x}$

(9.2) The system you need to solve using Lagrange multipliers is a system with $\underline{3}$ equations and $\underline{3}$ unknowns.

- (9.3) Using the method of Lagrange multipliers, write out the system of equations you need to solve for this problem. You **DO NOT** need to solve the system. Write your equations in the box below. (Do not just write the general equations.)

$$\begin{aligned} (1) \quad & 1 = \lambda(2x + 2y) \\ (2) \quad & 0 = \lambda(4y + 2x) \quad \text{OR} \quad \langle 1, 0 \rangle = \lambda \langle 2x + 2y, 4y + 2x \rangle \\ (3) \quad & x^2 + 2y^2 + 2xy - 8 = 0 \\ & \quad \quad \quad \text{or } x^2 + 2y^2 + 2xy = 8 \end{aligned}$$

Note: $\nabla f = \lambda \nabla g \Rightarrow \begin{cases} f_x = \lambda g_x \\ f_y = \lambda g_y \end{cases}$ must follow from what was given for $f(x, y)$ in 9.1 and the constraint equation in 9.3.

10. (10 pts) For each true/false question, fill in the appropriate bubble. No work required.

(10.1) ☒ (T) ☐ (F) For a function $f(x, y)$ and point $P = (a, b)$: $f_y(a, b) = D_{\mathbf{u}}f(a, b)$ when $\mathbf{u} = \langle 0, 1 \rangle$.

(10.2) ☐ (T) ☒ (F) If $f(x, y) = y \ln(x + y)$, then $(0, 0)$ is a critical point of f . *f not defined at (0,0)*

(10.3) ☒ (T) ☐ (F) The gradient $\nabla F(a, b, c)$ is perpendicular to the tangent plane to the level surface $F(x, y, z) = k$ at the point $P = (a, b, c)$.

(10.4) ☐ (T) ☒ (F) If a function $f(x, y)$ has continuous third-order partial derivatives then $f_{xxy} = f_{yyx}$.

(10.5) ☐ (T) ☒ (F) If $\nabla f(a, b) = \langle 0, 0 \rangle$, then f must have a local maximum or minimum at (a, b) .